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Transfer learning for materials informatics using crystal graph convolutional neural network

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ABSTRACT

For successful applications of machine learning in materials informatics, it is necessary to overcome the inaccuracy of predictions ascribed to insufficient amount of data. In this study, we propose a transfer learning using a crystal graph convolutional neural network (TL-CGCNN). Herein, TL-CGCNN is pretrained with big data such as formation energies for crystal structures, and then used for predicting target properties with relatively small data. We confirm that TL-CGCNN can improve predictions of various properties such as bulk moduli, dielectric constants, and quasiparticle band gaps, which are computationally demanding, to construct big data for materials. Moreover, we quantitatively observe that the prediction of properties in target models via TL-CGCNN becomes more accurate with an increase in size of training dataset in pretrained models. Finally, we confirm that TL-CGCNN is superior to other regression methods in the predictions of target properties, which suffer from small amount of data. Therefore, we conclude that TL-CGCNN is promising along with compiling big data for materials that are easy to accumulate and relevant to the target properties.

1. Introduction

To save time and human efforts, machine learning (ML) [1–6] has been widely applied to the materials informatics (MI) field, particularly for new material exploration. An important challenge in ML-based material exploration and design is the development of a generic descriptor (representation) that can be flexibly applied to predictions of multiple target properties. Because material properties depend on the crystal structure and constituent chemical elements, descriptors based on the structural features, such as Coulomb matrix [7,8], smooth overlap of atomic positions (SOAP) [9,10], and reciprocal 3D voxel space (R3DVS) [11], have been developed.

Recently, Xie and Grossman introduced a crystal graph convolutional neural network (CGCNN) [12,13]. As an input to a classification or regression model based on the CGCNN, only a crystal structure is necessary. The CGCNN constructs crystal graphs from crystal structures and predicts the target property using a deep neural network architecture. Numerous researchers have modified and improved on graph-based neural networks owing to their simplicity and flexibility. Park and Wolverton incorporated information of the Voronoi tessellated crystal structure into the CGCNN to encode three body correlations [14]. Sanyal *et al.* augmented the CGCNN with a multitask learning to predict multiple material properties simultaneously [15]. Chen *et al.* developed a MatErials Graph Network (MEGNet) that considers not only crystal structures but also molecules with additional attributes [16]. Karamad *et al.* implemented atomic orbital interaction features [17] and encoder-decoder networks in the CGCNN [18].

Another important challenge in ML-based material exploration and design is to overcome the unsatisfactory accuracy of predictive models because of insufficient amount of data [19]. Compared to other research fields such as computer vision [20] and natural language processing [21] which use big data comprising millions of data items [22], functional or practical properties in MI have not been substantially accumulated as their evaluation is computationally or experimentally demanding. Transfer learning (TL) [23] is a useful method for improving predictions suffering from insufficient amount of data.

TL is used to provide knowledge to other predictive models with different tasks (target variables). Herein, we focus on applications of TL for a target model with less amount of data by loading a pretrained model constructed with an enormous amount of data. In MI, studies have been conducted using TL. To classify Li superionic conductors, Cubuk *et al.* [24] constructed a large amount of labeled data based on

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crystal structure-based descriptors, and then reconstructed predictive models based on chemical element-based descriptors using a support vector machine. Yamada *et al.* [25] developed a pretrained model library called XenonPy.MDL [26] with big data for a multilayer neural network based on fingerprint descriptors [27], and it showed an improved prediction of organic or inorganic material properties by TL using the neural network. Jha *et al.* [28] demonstrated improved predictions of formation energies (ΔE_f) for smaller theoretical or experimental databases by TL using the ElemNet multilayer deep neural network [29], which uses elemental composition descriptors, and pretrained models for ΔE_f with a larger theoretical database [30,31].

Indeed, as deep convolutional neural networks become the mainstream in the computer vision field [23,32–35], TL is being increasingly used. The convolutional layers work as feature maps that can detect basic features such as vertical and horizontal lines, which are basic elements for recognizing objects. Li *et al.* [36] applied TL for microstructure reconstruction using deep convolutional neural networks pretrained with image big data such as ImageNet [22]. Based on the inspiration, CGCNN, on account of its neural network architecture, may have the potential for effective TL in MI.

In this study, we investigate TL using CGCNN (TL-CGCNN). Additionally, we quantitatively confirm that the predictions of basic material properties such as the Kohn–Sham band gap (E_g) and ΔE_f can be improved via TL-CGCNN. To confirm the flexible application of TL, we investigate whether TL-CGCNN can improve predictions of properties such as the bulk modulus (K_{VRH}) [37], dielectric constant (ε_r) [38], and quasiparticle band gap (GW-E_g) [39], which usually suffer from small amount of data and heavy computational costs. In addition, we discuss issues relevant to TL-CGCNN and its prospects.

2. Method

2.1. CGCNN

First, we review the background knowledge regarding the CGCNN constructed by Xie and Grossman [12,13]. Fig. 1 presents an overview of the CGCNN. The CGCNN involves the construction of graphs based on crystal structures and a deep neural network architecture including embedding, convolutional, pooling, and fully-connected (FC) layers.

The crystal structure is transformed into a multigraph G. Nodes and edges in the multigraph G indicate atoms and chemical bonds between the atoms, respectively. The multigraph G consists of a set of atoms in the crystal structure, undirected edges, atom features, and bond features. A bond feature vector ($\boldsymbol{u}_{(i,j)k}$) denotes the k-th bond between atoms i and j. An atom feature vector ($\boldsymbol{v}_i$) contains the properties of the i-th atom. The information of atoms and bonds in the multigraph G is prepared as discrete representations that consist of binary digits for describing constituent elements and bonds, such as group and periodic number of atoms, and atom distance. The discrete representations are transformed into continuous representations in the embedding layer. Thereafter, the continuous representations are input to the convolutional layer.

In the $(t+1)$-th convolutional layer,

$$\boldsymbol{v}_i^{(t+1)} = g\left[\left(\sum_{j,k} \boldsymbol{v}_j^{(t)} \oplus \boldsymbol{u}_{(i,j)_k}\right)\boldsymbol{W}_c^{(t)} + \boldsymbol{v}_i^{(t)}\boldsymbol{W}_s^{(t)} + \boldsymbol{b}^{(t)}\right], \tag{1}$$

where $\oplus$ denotes the concatenation of the atom and bond feature vectors of the neighboring atoms of the i-th atom. $\boldsymbol{W}_c^{(t)}$, $\boldsymbol{W}_s^{(t)}$, and $\boldsymbol{b}^{(t)}$ represent the optimized parameters in the neural network, namely, the convolution weight matrix, self-weight matrix, and bias of the t-th layer, respectively. $g(\cdot)$ denotes a non-linear activation function between layers, such as a softplus function [40].

However, because all neighbors in the convolution formulation share the weight matrix, different interactions of neighbors are not distinguished. Therefore, a standard edge-gating technique [41] can be applied for the formulation of a new type of convolution as follows: A neighbor feature vector $\boldsymbol{z}_{(i,j)k}$ is used as,

$$\boldsymbol{z}_{(i,j)k}^{(t)} = \boldsymbol{v}_i^{(t)} \oplus \boldsymbol{v}_j^{(t)} \oplus \boldsymbol{u}_{(i,j)k}, \tag{2}$$

and then,

$$\boldsymbol{v}_i^{(t+1)} = \boldsymbol{v}_i^{(t)} + \sum_{j,k} \sigma\left(\boldsymbol{z}_{(i,j)k}^{(t)}\boldsymbol{W}_c^{(t)} + \boldsymbol{b}_c^{(t)}\right) \circ g\left(\boldsymbol{z}_{(i,j)k}^{(t)}\boldsymbol{W}_s^{(t)} + \boldsymbol{b}_s^{(t)}\right), \tag{3}$$

where $\circ$ denotes element-wise multiplication (Hadamard product), and σ denotes a sigmoid function. Xie and Grossman [12] reported that the latter convolution formulation (Eq. (3)) showed better predictions compared to the former (Eq. (1)).

The atom feature vectors output by the convolutional layers are inserted into the pooling layer. Herein, an average pooling [42] is used. Thereafter, a crystal feature vector ($\boldsymbol{v}_g$) can be obtained by,

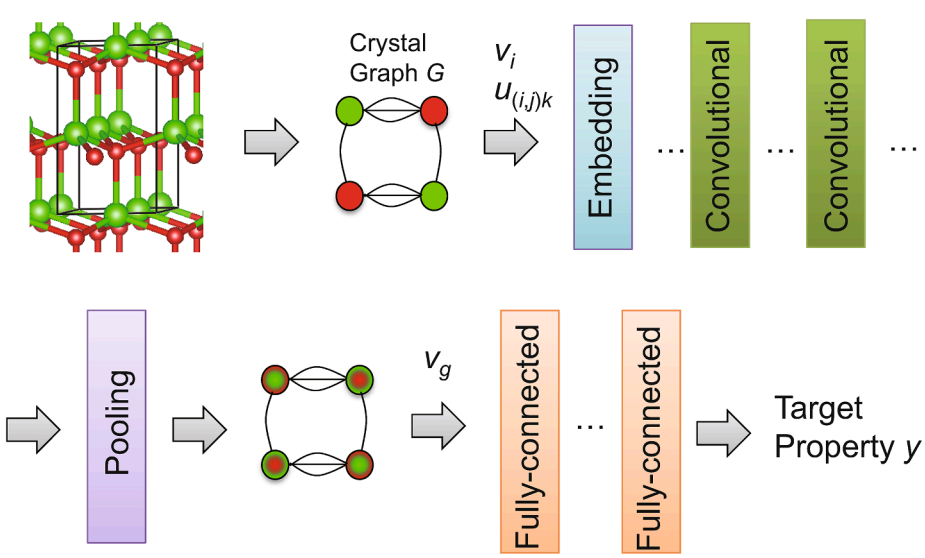

Fig. 1. Overview of the CGCNN. It predicts a target property with an input of the crystal structure, which is transformed into a crystal graph. Green and red atoms in the crystal structure correspond to the green and red nodes that are displayed as circles in the graph, respectively. The chemical bonds in the crystal structure correspond to the edges in the graph. The CGCNN architecture consists of embedding, convolutional, pooling, and FC layers.

$$\boldsymbol{v}_g = \frac{1}{N}\sum_i \boldsymbol{v}_i, \tag{4}$$

where N is the number of atoms in the crystal graph.

Finally, $\boldsymbol{v}_g$ is used as input to a network of FC layers with non-linearities that learn to predict a target property value, y, for the crystal. The predicted values can be obtained as follows:

$$\hat{y} = f\left(\boldsymbol{v}_g \boldsymbol{W}_g + \boldsymbol{b}_g\right), \tag{5}$$

where $\boldsymbol{W}_g$, $\boldsymbol{b}_g$, and $f(\cdot)$ are the weight matrix, bias, and non-linear function of the FC layers, respectively.

The $\boldsymbol{W}$ and $\boldsymbol{b}$ parameters are optimized to minimize the loss function, which is defined as the mean absolute difference between the predicted and target values for regressions, by a backpropagation procedure. Herein, stochastic gradient descent is employed as an optimizer [43].

2.2. TL-CGCNN

Fig. 2 presents a comparison of the concept of a conventional ML and a TL based on a convolutional neural network. For a conventional ML, we prepare each predictive model separately when we have different target variables. However, when we construct a "pretrained" (source) predictive model using big data based on a convolutional neural network in advance, an enormous number of parameters (weight and bias) in the convolutional layer can save the features of the big data. The parameters in the convolutional layer can be transferred to the target predictive model, which usually has a smaller amount of training data, as "knowledge".

There are two common methods in TL for treating the loaded parameters for convolutional neural networks. The first method is "fine tuning." Herein, the parameters of the entire layers of the pretrained model are initially loaded. Thereafter, all the parameters are optimized via backpropagation. This method is expected to have better optimization compared to when optimization is started with randomly generated initial parameters. To avoid losing the information of the pretrained model, it is recommended to select an adequate learning rate. The second method is "layer freezing." This method fixes the parameters of the earlier layers without additional optimizations by backpropagation. To save the information from the pretrained model, the optimization of parameters is only performed for the latter layers. In this study, we mainly used fine-tuning for TL-CGCNN.

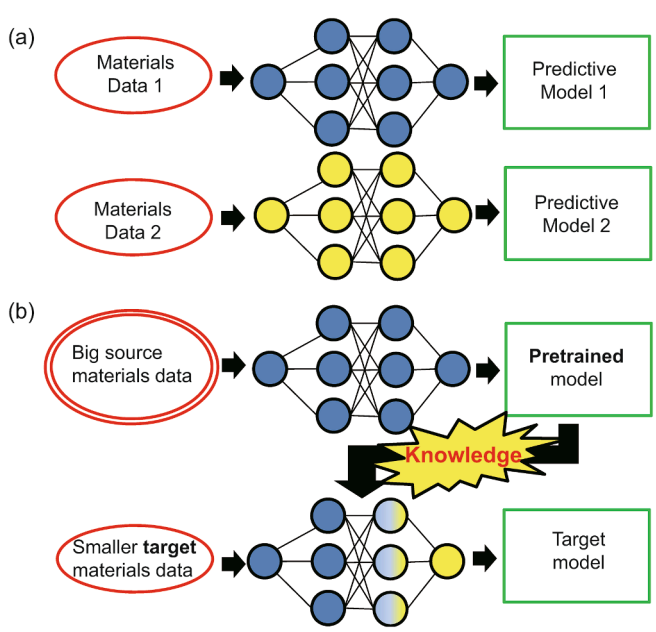

Fig. 2. Overview of TL. Comparison of (a) a conventional ML and (b) TL. For conventional ML, we prepare different predictive models separately when we have different tasks. However, when using TL, the knowledge (parameters, namely, weights and biases) accumulated in the pretrained model with big data can be transferred to the target predictive model.

In addition, to match the scales of target properties between the pretrained and target models in this study, tanh-normalization [44] was used for the predictive models as follows:

$$y' = \frac{1}{2}\left\{\tanh\left(\frac{0.01(y-\mu)}{s}\right)+1\right\}, \tag{6}$$

where tanh, μ, and s are the hypertangent function, and mean and standard deviation of the target variable, y, respectively. After the predictive model is optimized, it is denormalized when the prediction error is summarized.

2.3. Data and evaluation of predictive model

Crystal structures and their corresponding E_g and ΔE_f (defined with respect to the most stable state of each elemental standard state) have been compiled from the first-principles calculations database, Materials Project Database (MPD) [45]. They were calculated based on the exchange–correlation function of generalized gradient approximation (GGA) parameterized by Perdew-Burke-Ernzerhof (PBE) form [46] in the Vienna *Ab-initio* Simulation Package (VASP) [47,48]. The on-site Coulomb interactions were treated using GGA + U in case transition metals were included in the materials [49]. The effective U values for the transition metals are summarized in Supplementary Table S3. We used 118,286 computed data samples (assessed on November 11th, 2019) that are available in MPD to construct crystal graphs with sufficient connectivity between atoms. Within a cutoff radius of 8 Å [13], twelve nearest neighboring atoms were considered to construct the crystal graphs [12]. Crystal graphs with insufficient connectivity, which do not have twelve neighboring atoms within the cutoff radius, were not included in the data. He, Ar, and Ne were also not included in the data. Among them, 59,923 data samples were used as nonmetallic (NM) materials under the condition, computed E_g greater than 0.1 eV. The CGCNN was used to predict the ΔE_f value with the input of only crystal structures for all materials. For these NM materials, the CGCNN was used to predict the E_g as well as the ΔE_f value.

In this study, large sizes of datasets, namely, 10000, 54000, and 113,000 are written as 10 k, 54 k, and 113 k, respectively. The ratio of size of training, validation, and test datasets was set to 4:1:1 for data selection of the CGCNN. However, the sizes of the validation and test datasets were fixed as 2500 when the size of the training dataset was greater than 10 k. Weights and biases were optimized to minimize the loss function by using training data. Validation data were used to find the best epoch to stop the backpropagation with the minimum loss function, which indicates the threshold point of overfitting. Test data that did not participate in the modeling were used to evaluate the prediction error. In this study, the mean absolute error (MAE) of the test data was used to evaluate the prediction error. The prediction errors were evaluated by averaging over several trials with different datasets. When the size of the training dataset was 500, greater than 500 and less than or equal to 10 k, and greater than 10 k, the number of trials was set to 10, 5, and 2, respectively. We used random selections of training/validation/test data based on random seeds. Therefore, although training/validation/test data vary among different trials, the same type of data could be used for comparisons of various predictive models when the same amount of data is used. The maximum iteration of epochs was mostly set to 500, ensuring that the minimum MAE of the validation data was already obtained at an epoch less than 500. When this was not satisfied, we increased the maximum iteration of epochs until we obtained the minimum MAE of the validation data.

TL-CGCNNs were employed to predict E_g or ΔE_f with less data using

the pretrained CGCNN models that were constructed to predict E_g or ΔE_f with 10 k, 54 k, and 113 k (only available for ΔE_f) training data. For the target models, the ratio of the size of training, validation, and test datasets was also set to 4:1:1.

To investigate the effect of the TL-CGCNN on more practical cases that suffer from insufficient amount of data, we used the target properties of K_{VRH}, ε_r, and GW-E_g. The K_{VRH} [37] and ε_r [38] values were obtained from MPD. For the K_{VRH}, we prepared 12,990 data samples with values ranging from 2 GPa (Cs metal) to 436 GPa (diamonds), and also obtained by the Voigt-Reuss-Hill approximation [50]. For the ε_r, we prepared 4182 NM data samples (with a condition of $E_g > 0.1$ eV). The ε_r value is prepared with an assumption to be a single scalar polycrystalline value converted from a dielectric tensor [38]. The GW-E_g values were obtained from a previous study [39]. To obtain the GW-E_g values, the G_0W_0 calculations [51] with Heyd-Scuseria-Ernzerhof hybrid functional (G_0W_0@HSE06) [52] were used for 270 binary and ternary materials (inorganic compounds) with optimized crystal structures by PBE-sol functional [53].

Supplementary Table S1 summarizes the main hyperparameters that were used for the CGCNN models in this study. The default values of other hyperparameters can be referred to in the original code [13]. Among various hyperparameters, the length of the feature vector of the FC layer and the initial learning rate of the target model were reduced when we used TL-CGCNNs.

We compared the prediction of the TL-CGCNN with other widely used regression methods for various predictive models. Partial least squares (PLS) [54], least absolute shrinkage and selection operator (LASSO) [55], support vector regression (SVR) [56], and random forests (RF) [57] were employed as the regression methods. As descriptors for these regression methods, a generalized set of chemical element-based descriptors (132 descriptors) [2] obtained by the matminer [58] Python library were used. For the SVR, a radial basis function (RBF) kernel was used. In addition, SOAP kernel [9,10] was employed as a representative structural descriptor-based predictive model. DScribe package [59] was used to calculate the SOAP of each atom and construct SOAP kernels formed by a pair of local environments. Ridge regression [60] with SOAP kernel descriptors was used to predict the properties. The sizes of training + validation and test datasets were 625 and 125, respectively. For training and validation data, 5-fold cross validations were used to optimize the hyperparameters relevant to the minimization of overfitting for the regression methods.

3. Results and discussion

3.1. CGCNN

Herein, before describing the prediction results, we note that the names of the pretrained and target models are written based on the size of the training dataset and target property. For instance, 500-NM-E_g indicates that the target property of the predictive model is E_g, the size of the training dataset is 500 (sizes of both validation and test datasets are 125), and only NM data are included in the datasets.

Table 1 summarizes material-property datasets that are used in this study. Most of the materials are bulk-type inorganic materials, and their crystal structures and properties are loaded from the MPD. A mean absolute difference of the data indicates a baseline for a predictive model.

All the data were separated into NM and metallic data using E_g with a threshold value of 0.1 eV. However, when we consider constituent chemical elements in the materials, NM (semiconductors and insulators) materials, such as oxides, are mainly inorganic compounds formed by combining cationic and anionic elements, whereas metallic materials, such as transition metal alloys, are mainly alloys consisting of combinations of metallic elements. For instance, among 59,923 NM materials, 52,089 are oxides, nitrides, and halogenides. The NM materials mainly have ionic or covalent bonds, whereas the metallic materials mainly have metallic bonds. Therefore, we consider that the atom and bond features saved in the neural network architecture may be different from each other.

Fig. 3 shows scatter plots between the ΔE_f and E_g properties of 118,286 materials. The scatter plots indicate that there is no strong correlation between ΔE_f and E_g. The linear correlation coefficients (r_p) between the two properties for the entire data and NM data are −0.49 and −0.33, respectively. For the 59,923 NM materials, the MAE of the ΔE_f and E_g obtained via an ordinary linear regression using E_g and ΔE_f

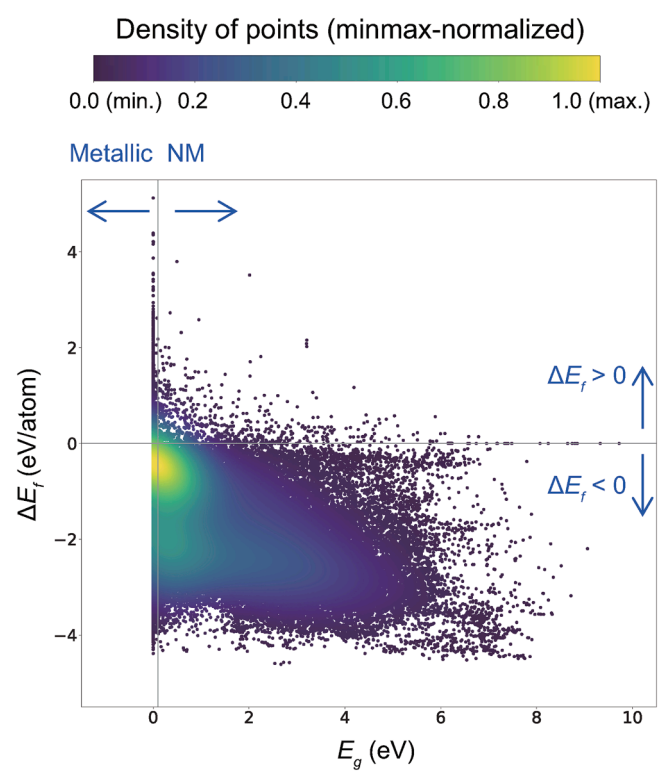

Fig. 3. Scatter plots between the ΔE_f and E_g quantities of 118,286 materials. The color scale represents the relative density of points generated via Gaussian kernel density estimation [61]. The probability density function is minmax-normalized, and the Gaussian bandwidth is three times the Scott's factor [61].

Table 1
Summary of the material-property datasets used in this study.

Material Property	Size of dataset	Minimum value	Maximum value	Mean value	Mean absolute difference	Data source Ref.
E_g	59,923	0.100 eV	9.721 eV	2.179 eV	1.256 eV	[45]
ΔE_f	118,286	−4.612 eV/atom	5.124 eV/atom	−1.481 eV/atom	0.968 eV/atom	[45]
ΔE_f (NM)	59,923	−4.612 eV/atom	3.797 eV/atom	−2.021 eV/atom	0.768 eV/atom	[45]
K_{VRH}[a,b]	12,990	0.301	2.639	1.874	0.304	[45]
ε_r[a]	4182	0.063	2.992	1.096	0.265	[45]
GW-E_g	270	0.357 eV	14.548 eV	4.996 eV	2.486 eV	[39]

[a] The values are in $\log_{10}$.
[b] The unit in $\log_{10}$ is GPa.

were 0.72 eV/atom and 1.17 eV, respectively. This implies that more complex regression methods and descriptor sets are necessary for their predictions.

Fig. 4 shows the dependence of the prediction error of the test data by CGCNN for E_g and ΔE_f on the size of training dataset. The prediction errors decrease with an increase in the size of training dataset. When the sizes of training datasets are 54 k and 113 k, the prediction errors for E_g and ΔE_f are 0.358 eV and 0.046 eV/atom, respectively. The prediction error of ΔE_f is comparable to the accuracy of the density functional theory (DFT) method with respect to the experiments [31]. We expect that the prediction error can further decrease when we collect more training data. This is because the decreasing tendency is not saturated yet. In addition, both prediction error and size of training dataset in $\log_{10}$ scale shows a strong linear relationship, as reported in previous ML models for other systems [62,63]. The r_p of E_g, ΔE_f, and NM-ΔE_f vs. size of training dataset in $\log_{10}$ scale are −1.000, −0.999, and −1.000, respectively.

As mentioned above, because the bond natures and constituent elements of the NM and metallic materials are different, predictive models for ΔE_f, with only NM material data were separately constructed. The prediction error with only NM material data is smaller than that with metallic and NM material data combined when the same size of training dataset is used.

3.2. CGCNN-based TL-CGCNN

Predictive models usually suffer from unsatisfactory accuracy owing to a small amount of data. Therefore, we intentionally prepared target models with a small amount of data for applying a TL-CGCNN to E_g and ΔE_f. The results are shown in Fig. 5, and numerical values are summarized in Supplementary Table 2. The pretrained models with a large amount of data were used for predicting targets of counterpart properties.

First, Fig. 5(a) shows a comparison of the predictive performances for E_g. When the size of the training dataset is 500, the prediction error of E_g by the CGCNN without TL is 0.978 eV. Using TL-CGCNN with the 10 k-ΔE_f, 54 k-ΔE_f, and 113 k-ΔE_f pretrained models, the prediction error decreases to 0.866, 0.826, and 0.815 eV, respectively. The more training data we use for the pretrained models, the more the prediction powers are improved.

In addition, we performed a statistical significance analysis to judge whether the decrease in prediction error via TL is meaningful or accidental. For this analysis, the paired *t*-test was employed. To confirm that the mean averages of the two distributions are different, the *p*-value should be lower than 0.01 (threshold for 99% significance level). The *p*-value of the paired *t*-test using ten MAE values from the CGCNN and TL-CGCNN with the 113 k-ΔE_f pretrained model is 6.9×10^{-5}. Therefore, we conclude that the TL-CGCNN exhibits better performance than the CGCNN with a 99% significance level for this prediction.

Second, we investigated predictive performances for ΔE_f. The prediction error of ΔE_f by the CGCNN without TL is 0.239 eV/atom when the size of the training dataset is 500. When we use TL-CGCNNs with the 10 k-NM-E_g and 54 k-NM-E_g pretrained models, the prediction error decreases to 0.236 and 0.222 eV/atom, respectively. However, despite the small decrease in prediction error, we conclude that an improvement via the TL is not achieved. This is because the *p*-value of the statistical significance analysis between the CGCNN and TL-CGCNN with the 54 k-NM-E_g pretrained model is too large (0.15).

Additionally, Fig. 5(b) shows a comparison of the predictive performances for ΔE_f, but with only NM material data (NM-ΔE_f). When the target model is 500-NM-ΔE_f, the prediction error by the TL-CGCNN with the 54 k-NM-E_g pretrained model is 0.149 eV/atom, which is much smaller than that by the CGCNN without TL (0.184 eV/atom). Contrary to the former predictive models (ΔE_f of all data), the *p*-value of the statistical significance analysis between the CGCNN and TL-CGCNN with the 54 k-NM-E_g pretrained model is small (4×10^{-4}). Therefore, we conclude that the TL-CGCNN exhibits better performance than the CGCNN with a 99% significance level for this prediction. In this case, the data of both pretrained and target models were NM data. The bond features are mainly formed by ionic or covalent bonds, and the crystal graphs are formed by combinations of cations and anions. We presume that this pretrained information could be smoothly transformed into the predictions for the NM material data, but not for all data, such as the data for metallic materials, which are less overlapped with the pretrained data in the descriptor space.

Subsequently, we tested the dependence of the TL-CGCNN on the size of dataset of the target models. Using the 113 k-ΔE_f pretrained model, the prediction errors are 0.866, 0.752, 0.553, and 0.490 eV when the target models are 500-E_g, 1000-E_g, 5000-E_g, and 10 k-E_g, respectively. As shown in Fig. 5(c), the percentage improvements in prediction error for the case without TL tend to decrease by 16.7, 12.0, 9.0 and 3.2%, respectively. This was also tested for the prediction of ΔE_f for the NM

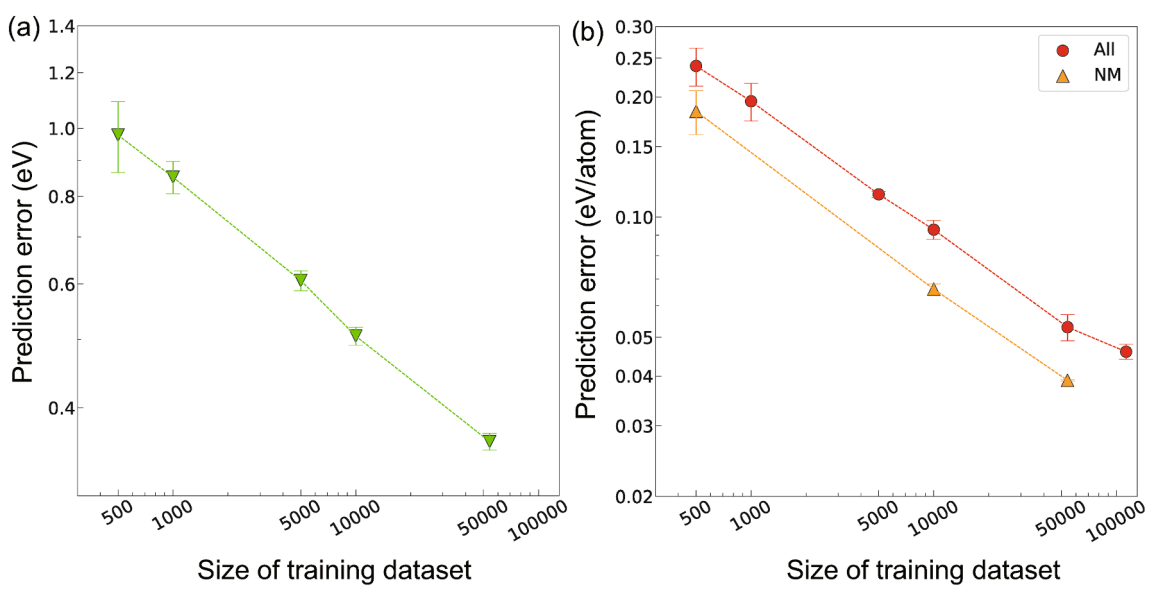

Fig. 4. Dependence of prediction error (MAE of test data) for (a) E_g and (b) ΔE_f on size of the training dataset by CGCNN. Horizontal and vertical axes are in $\log_{10}$ scale. Error bars indicate one standard deviation for prediction errors from several predictive models with different data samplings.

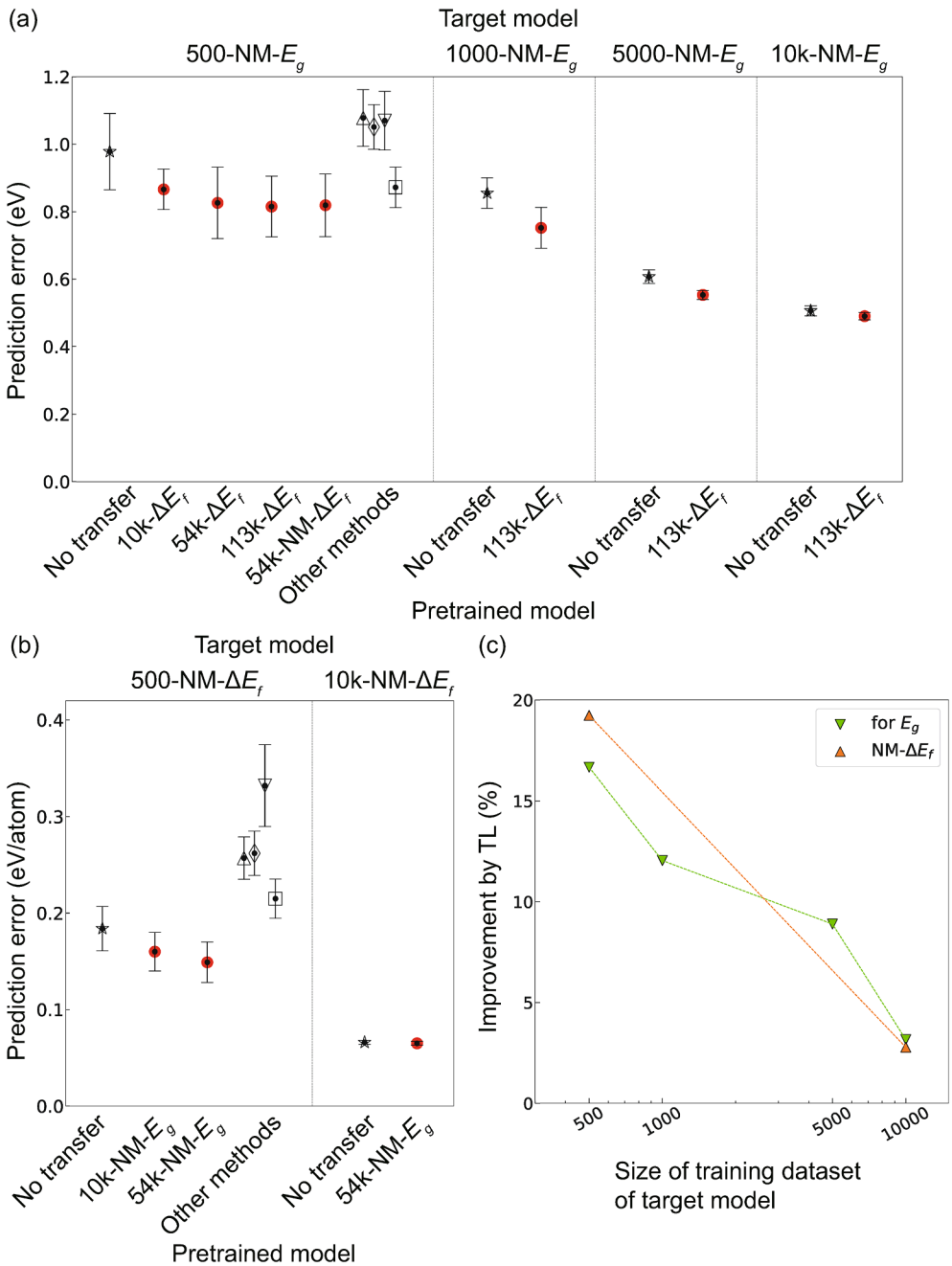

Fig. 5. Comparison of prediction error for (a) E_g and (b) NM-ΔE_f. The predictions via the CGCNN (no transfer) and TL-CGCNN are displayed as open star and red circle symbols, respectively. The predictions via other regression methods, namely, PLS, LASSO, SVR, and RF, with elemental descriptors are displayed as open up-pointing triangle, diamond, down-pointing triangle, and square symbols, respectively. Error bars indicate one standard deviation for prediction errors from ten predictive models with different data samplings. The baselines, obtained as the mean absolute differences of the data distribution, for E_g and NM-ΔE_f are 1.26 and 0.77 (in the same unit as in each subfigure), respectively. (c) Dependence of percentage improvement of prediction error by best TL-CGCNN with respect to plain CGCNN on size of training dataset of target model.

data. Using the 54 k-NM-E_g pretrained model, the prediction errors are 0.149 and 0.065 eV/atom (19.2 and 2.8% improvement in predictions without the TL-CGCNN) when the target models are 500-NM-ΔE_f and 10 k-NM-ΔE_f, respectively. This implies that using the TL-CGCNN is more efficient when we have less data.

We confirmed that the TL-CGCNN for E_g and ΔE_f with pretrained models based on the counterpart properties improved the predictions despite a weak correlation, as shown in Fig. 3. This implies that the atom and bond features from crystal graphs captured in the convolutional layers in the pretrained model with larger data are flexibly adapted to predict other material properties.

3.3. TL-CGCNN for computationally demanding properties

Additionally, we tested the TL-CGCNN for predictive models for other properties, namely, K_{VRH}, ε_r, and GW-E_g. Because these properties are more computationally expensive, there are much less data than E_g or ΔE_f data. Considering their broad distributions, the K_{VRH} and ε_r values in a $\log_{10}$ scale were used. The unit of the K_{VRH} value inside the $\log_{10}$ is GPa.

Fig. 6 shows the prediction errors of three properties via the CGCNN without TL and TL-CGCNN. The prediction error of the 500-K_{VRH} by the CGCNN without TL is 0.123. When the 113 k-ΔE_f pretrained model is used, the prediction error decreases to 0.112 (8.7%). The prediction error of the 500-NM-ε_r by the CGCNN without TL is 0.181. It decreases to 0.169 (7.1%) and 0.163 (10.0%) when the 113 k-ΔE_f and 54 k-NM-E_g pretrained models are employed, respectively. Despite the additional training data of the 113 k-ΔE_f pretrained model, a smaller prediction error of the 54 k-NM-E_g pretrained model-based prediction is obtained. For 4182 NM materials that have ε_r data, this may be ascribed to the correlation (−0.48) between E_g and ε_r (in $\log_{10}$ scale) being much stronger than the correlation (0.07) between ΔE_f and ε_r (in $\log_{10}$ scale).

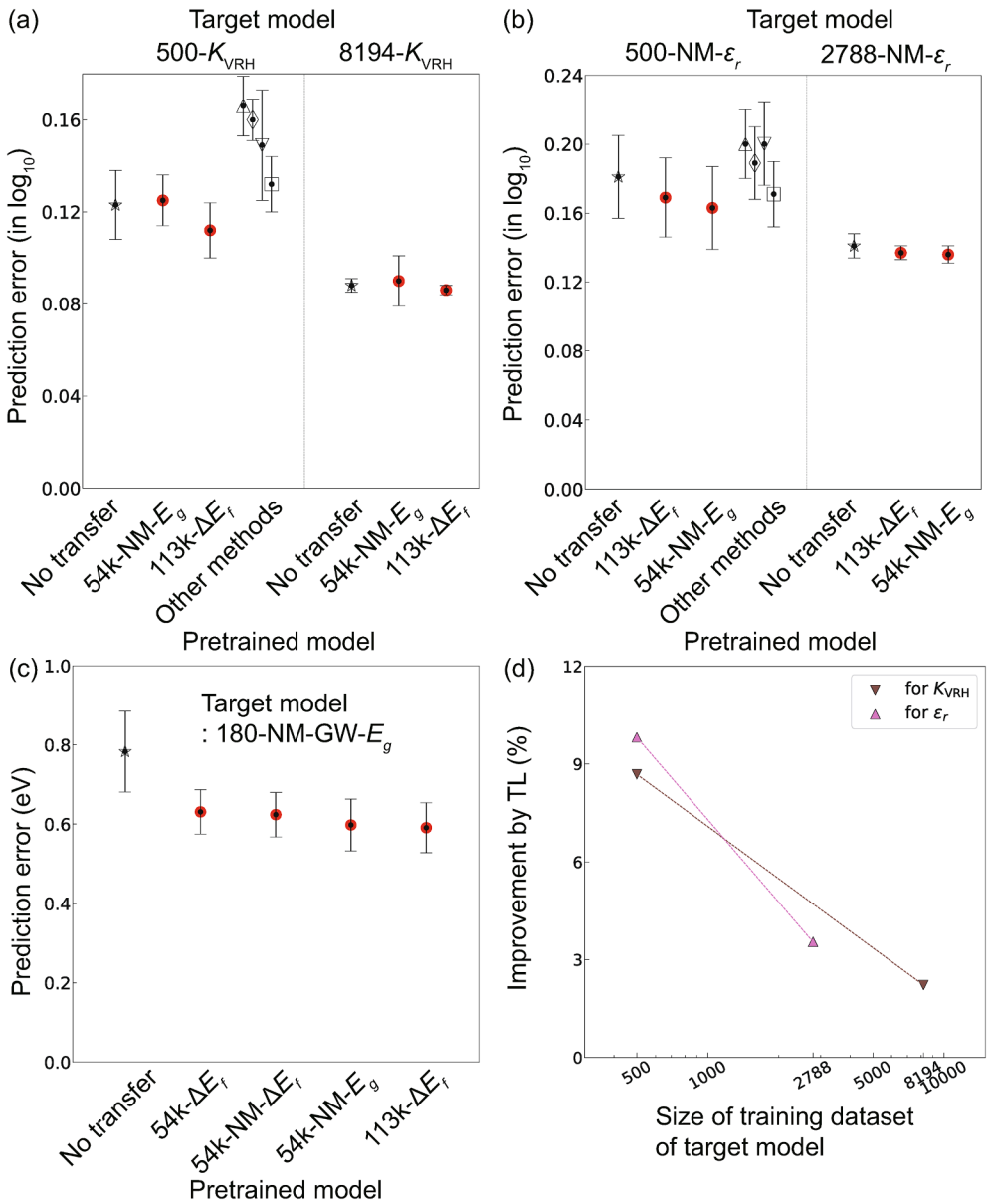

Fig. 6. Comparison of prediction errors of (a) K_{VRH}, (b) ε_r, and (c) GW-E_g. The predictions via the CGCNN (no transfer) and TL-CGCNN are displayed as open star and red circle symbols, respectively. The predictions via other regression methods, namely, PLS, LASSO, SVR, and RF, with elemental descriptors are displayed as open up-pointing triangle, diamond, down-pointing triangle, and square symbols, respectively. The unit of K_{VRH} in $\log_{10}$ is GPa. Error bars indicate one standard deviation for prediction errors from ten predictive models with different data samplings. The baselines, obtained as the mean absolute differences of the data distribution, for K_{VRH}, ε_r, and GW-E_g are 0.30, 0.27, and 2.49 (in the same unit as in each subfigure), respectively. (d) Dependence of percentage improvement of prediction error by best TL-CGCNN with respect to plain CGCNN on size of training dataset of target model.

The prediction error of the 180-NM-GW-E_g by the CGCNN without TL is 0.783 eV. The smallest prediction error by the TL-CGCNN among various pretrained models is achieved by the 113 k-ΔE_f pretrained model, whose value decreases to 0.591 eV (24.5%). For the 500-K_{VRH}, 500-NM-ε_r, and 180-NM-GW-E_g predictions, the p-values obtained via the statistical significance analysis between the CGCNN without TL and corresponding best TL-CGCNN models are 0.002, 0.004, and 2×10^{-4}, respectively. Therefore, we conclude that the TL-CGCNN shows better performance than the CGCNN without TL for these three predictions with a 99% significance level.

When more data are used for the prediction of K_{VRH} and ε_r, (8194 and 2788 training data for K_{VRH} and ε_r, respectively), we obtain a decreased prediction error by the TL-CGCNN. However, the improvement via TL tends to decrease compared to when the target model has less data as shown in Fig. 6(d). The percentage improvements in K_{VRH} and ε_r using TL are 1.7 and 3.7%, respectively. This implies that the TL-CGCNN may be more useful for predictions of target properties that suffer from small amount of data.

Table 2 shows equivalent size of training dataset for CGCNN to achieve the same prediction error by corresponding best TL-CGCNN with 500 training data of target model. This is obtained by an interpolation using the linear relationship between $\log_{10}$(prediction error) and $\log_{10}$(size of dataset) as shown in Fig. 4(b). The equivalent sizes of training dataset for CGCNN to achieve the same prediction error by corresponding best TL-CGCNN become approximately twice. Accordingly, these results demonstrate that using the TL-CGCNN is useful for such practical properties that usually suffer from limited size of dataset.

Table 2
Equivalent size of training dataset for CGCNN to achieve the same prediction error obtained by corresponding best TL-CGCNN when the size of the training dataset of target model is 500.

Target model	500-E_g	500-NM-ΔE_f	500-K_{VRH}	500-NM-ε_r
Equivalent size of training dataset	1200	925	1068	1028

3.4. Additional issues

We also tested predictions of 500-NM-E_g, 500-NM-ΔE_f, 500-K_{VRH}, and 500-NM-ε_r via PLS [54] LASSO [55], SVR [56], and RF [57]. These methods were supervised using a generalized set of chemical element-based descriptors [2]. The results are shown in Figs. 5 and 6. In addition, there are (similarity) kernel-based models [9,10,63–65] that use structural descriptors based on information of atomic coordinates. In this study, SOAP kernel [9,10] with ridge regression [60] was used to compare the predictions. The numerical values of the predictive performances are summarized in Supplementary Table S2.

Among the four regression methods with the chemical element-based descriptors, RF exhibits the best performance. The CGCNN exhibits better performance in the predictions of 500-NM-ΔE_f and 500-K_{VRH}, but poorer performance in the predictions of 500-NM-E_g and 500-NM-ε_r than the RF. However, the TL-CGCNN achieves the smallest prediction errors for the four predictions. The predictions with other widely used regression methods such as RF can be improved with descriptors that are more specialized and have strong correlations with the target properties; however, it is not easy to find additional useful descriptors for each target property.

Additionally, as a descriptor of local structure, we employed SOAP that has been extensively used for regression models or generating ML potentials [66,67]. For organic molecules [10,65,68] or nanoclusters [69–71], the target properties can be efficiently supervised by structural descriptors such as bond lengths and angles. This is because most of the data consist of limited types of chemical elements such as carbon, hydrogen, oxygen, and nitrogen. On the contrary, the MPD data used in this study comprise various inorganic materials; therefore, most of the chemical elements in the periodic table participate in the data. Notably, as summarized in Supplementary Table S2, the SOAP kernel did not show better performance compared to the TL-CGCNN. The prediction errors by the SOAP kernel for predictions of 500-NM-E_g, 500-NM-ΔE_f, 500-K_{VRH}, and 500-NM-ε_r are larger compared to those by the RF with the chemical element-based descriptors. This indicates that there is little similarity in local environments among chemical elements in such a small number (five hundred) of data samples selected from a wide range of inorganic materials. Karamad *et al.* [18] reported that SOAP descriptors with kernel ridge regression showed similar performance to that of the CGCNN for predictions of ΔE_f and E_g of inorganic materials using more than 26 k data samples from MPD.

Therefore, we emphasize that the TL-CGCNN is a powerful and flexible prediction tool with merits for a small amount of data. This tool facilitates the construction of pretrained models with big data, and it can systematically capture features of both chemical elements and crystal structures.

Deep neural networks such as CGCNN have many hyperparameters. For an effective TL, we need to carefully select hyperparameters based on the following considerations: transferred information is to be conserved and flexibly used for the target predictive model, and overfitting is to be reduced. Based on these considerations, we optimized the learning rate and the length of the vector of the FC layer for a case study. The results are described in Supplementary Chapter S1.

Fine-tuning has been used as the TL-CGCNN method in this study. Additionally, we tested the layer freezing method. We only optimized the parameters of the last FC layer, freezing those of the earlier layers. As a result, for the investigated prediction targets, fine-tuning exhibits better predictions. The detailed results are described in Supplementary Chapter S2. In some previous studies using deep convolutional neural networks with many multiple layers [23,33], the concept of mixing fine-tuning and layer freezing has been suggested, that is, only early convolutional layers are frozen and the other layers including convolutional layers are optimized.

4. Conclusion

In this study, TL-CGCNNs were applied to the predictive models of various material properties, which suffer from inaccuracy owing to small datasets of target properties. We quantitatively confirmed that the predictions can be improved using the TL-CGCNN with the pretrained model constructed with big data. In addition, we quantitatively observed that the prediction of properties in target models via TL-CGCNN becomes more accurate when the pretrained model is trained with bigger data and the pretrained and target models are more strongly correlated. Because the problem of an insufficient amount of data arises in many MI issues, we expect that a combination of general descriptors such as CGCNN and TL can be more widely employed.

CRediT authorship contribution statement

Joohwi Lee: Investigation, Methodology, Visualization, Software, Writing - original draft, Writing - review & editing. **Ryoji Asahi:** Conceptualization, Investigation, Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.commatsci.2021.110314.

References

[1] B. Meredig, A. Agrawal, S. Kirklin, J.E. Saal, J. Doak, A. Thompson, K. Zhang, A. Choudhary, C. Wolverton, Phys. Rev. B 89 (2014), 094104.
[2] L. Ward, A. Agrawal, A. Choudhary, C. Wolverton, npj Comput. Mater. 2 (2016) 16028.
[3] R. Ramprasad, R. Batra, G. Pilania, A. Mannodi-Kanakkithodi, C. Kim, npj Comput. Mater. 3 (2017) 54.
[4] K.T. Butler, D.W. Davies, H. Cartwright, O. Isayev, A. Walsh, Nature 559 (2018) 547.
[5] J. Schmidt, M.R. Marques, S. Botti, M.A. Marques, npj Comput. Mater. 5 (2019) 83.
[6] M. Haghighatlari, J. Hachmann, Curr. Opin. Chem. Eng. 23 (2019) 51.
[7] M. Rupp, A. Tkatchenko, K.-R. Müller, O.A. Von Lilienfeld, Phys. Rev. Lett. 108 (2012), 058301.
[8] G. Montavon, M. Rupp, V. Gobre, A. Vazquez-Mayagoitia, K. Hansen, A. Tkatchenko, K.-R. Müller, O.A. Von Lilienfeld, New J. Phys. 15 (2013), 095003.
[9] A.P. Bartók, R. Kondor, G. Csányi, Phys. Rev. B 87 (2013), 184115.
[10] S. De, A.P. Bartók, G. Csányi, M. Ceriotti, Phys. Chem. Chem. Phys. 18 (2016) 13754.
[11] S. Kajita, N. Ohba, R. Jinnouchi, R. Asahi, Sci. Rep. 7 (2017) 16991.
[12] T. Xie, J.C. Grossman, Phys. Rev. Lett. 120 (2018), 145301.
[13] CGCNN (Compatible with Pytorch Ver. 0.3.1), <https://github.com/txie-93/cgcnn> (accessed Feb. 20, 2019).
[14] C.W. Park, C. Wolverton, Preprint at <https://arxiv.org/abs/1906.05267> (2019).
[15] S. Sanyal, J. Balachandran, N. Yadati, A. Kumar, P. Rajagopalan, S. Sanyal, P. Talukdar, Preprint at <https://arxiv.org/abs/1811.05660> (2018).
[16] C. Chen, W. Ye, Y. Zuo, C. Zheng, S.P. Ong, Chem. Mater. 31 (2019) 3564.
[17] T. Lam Pham, H. Kino, K. Terakura, T. Miyake, K. Tsuda, I. Takigawa, H. Chi Dam, Sci. Tech,, Adv. Mater. 18 (2017) 756.
[18] M. Karamad, R. Magar, Y. Shi, S. Siahrostami, I.D. Gates, A.B. Farimani, Phys. Rev. Mater. 4 (2020), 093801.
[19] Y. Zhang, C. Ling, npj Comput. Mater. 4 (2018) 25.
[20] Y. LeCun, Y. Bengio, G. Hinton, Nature 521 (2015) 436.
[21] T. Young, D. Hazarika, S. Poria, E. Cambria, IEEE Comput. Intell. Mag. 13 (2018) 55.

[22] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, in: 2009 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Miami, FL, 2009, pp. 248–255.
[23] J. Yosinski, J. Clune, Y. Bengio, H. Lipson, in: Advances in Neural Information Processing Systems, Curran Associates Inc, Montréal, Canada, 2014, pp. 3320–3328.
[24] E.D. Cubuk, A.D. Sendek, E.J. Reed, J. Chem. Phys. 150 (2019), 214701.
[25] H. Yamada, C. Liu, S. Wu, Y. Koyama, S. Ju, J. Shiomi, J. Morikawa, R. Yoshida, ACS Cent Sci. 5 (2019) 1717.
[26] XenonPy, <https://github.com/yoshida-lab/xenonpy>. (accessed Feb. 18, 2020).
[27] RDKit: Open-Source cheminformatics software. <http://www.rdkit.org>. (accessed Feb. 18, 2020).
[28] D. Jha, K. Choudhary, F. Tavazza, W.-K. Liao, A. Choudhary, C. Campbell, A. Agrawal, Nat. Commun. 10 (2019) 5316.
[29] D. Jha, L. Ward, A. Paul, W.-K. Liao, A. Choudhary, C. Wolverton, A. Agrawal, Sci. Rep. 8 (2018) 17593.
[30] J.E. Saal, S. Kirklin, M. Aykol, B. Meredig, C. Wolverton, JOM 65 (2013) 1501.
[31] S. Kirklin, J.E. Saal, B. Meredig, A. Thompson, J.W. Doak, M. Aykol, S. Rühl, C. Wolverton, npj Comput. Mater. 1 (2015) 15010.
[32] J. Hoffman, S. Guadarrama, E.S. Tzeng, R. Hu, J. Donahue, R. Girshick, T. Darrell, K. Saenko, in: Advances in Neural Information Processing Systems, Curran Associates Inc, Montréal, Canada, 2014, pp. 3536–3544.
[33] D. Soekhoe, P. Van Der Putten, A. Plaat, in: 15th International Symposium on Intelligent Data Analysis, Springer, Stockholm, Sweden, 2016, pp. 50–60.
[34] K. Gopalakrishnan, S.K. Khaitan, A. Choudhary, A. Agrawal, Constr. Build. Mater. 157 (2017) 322.
[35] A. Kensert, P.J. Harrison, O. Spjuth, SLAS Discov. 24 (2019) 466.
[36] X. Li, Y. Zhang, H. Zhao, C. Burkhart, L.C. Brinson, W. Chen, Sci. Rep. 8 (2018) 13461.
[37] M. De Jong, W. Chen, T. Angsten, A. Jain, R. Notestine, A. Gamst, M. Sluiter, C. K. Ande, S. Van Der Zwaag, J.J. Plata, Sci. Data 2 (2015), 150009.
[38] I. Petousis, D. Mrdjenovich, E. Ballouz, M. Liu, D. Winston, W. Chen, T. Graf, T. D. Schladt, K.A. Persson, F.B. Prinz, Sci. Data 4 (2017), 160134.
[39] J. Lee, A. Seko, K. Shitara, K. Nakayama, I. Tanaka, Phys. Rev. B 93 (2016), 115104.
[40] C. Dugas, Y. Bengio, F. Bélisle, C. Nadeau, R. Garcia, in: Advances in Neural Information Processing Systems, MIT Press, Denver, CL, 2000, pp. 472–478.
[41] D. Marcheggiani, I. Titov, in: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Copenhagen, Denmark, 2017, pp. 1506–1515.
[42] D.K. Duvenaud, D. Maclaurin, J. Iparraguirre, R. Bombarell, T. Hirzel, A. Aspuru-Guzik, R.P. Adams, in: Advances in neural information processing systems, Curran Associates Inc, Montréal, Canada, 2015, pp. 2224–2232.
[43] L. Bottou, O. Bousquet, in: Advances in neural information processing systems, Curran Associates Inc, Vancouver, Canada, 2007, pp. 161–168.
[44] A. Jain, K. Nandakumar, A. Ross, Pattern Recogn. 38 (2005) 2270.
[45] A. Jain, S.P. Ong, G. Hautier, W. Chen, W.D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, K.A. Persson, Appl. Phys. Lett. Mater. 1 (2013), 011002.
[46] J.P. Perdew, K. Burke, M. Ernzerhof, Phys. Rev. Lett. 77 (1996) 3865.
[47] G. Kresse, J. Furthmüller, Comput. Mater. Sci. 6 (1996) 15.
[48] G. Kresse, J. Furthmüller, Phys. Rev. B 54 (1996) 11169.
[49] S.L. Dudarev, G.A. Botton, S.Y. Savrasov, C.J. Humphreys, A.P. Sutton, Phys. Rev. B 57 (1998) 1505.
[50] D. Chung, W. Buessem, J. Appl. Phys. 38 (1967) 2535.
[51] F. Fuchs, J. Furthmüller, F. Bechstedt, M. Shishkin, G. Kresse, Phys. Rev. B 76 (2007), 115109.
[52] J. Heyd, G.E. Scuseria, M. Ernzerhof, J. Chem. Phys. 124 (2006), 219906.
[53] J. Perdew, A. Ruzsinszky, G. Csonka, O. Vydrov, G. Scuseria, L. Constantin, X. Zhou, K. Burke, Phys. Rev. Lett. 100 (2008), 136406.
[54] P. Geladi, B.R. Kowalski, Anal. Chim. Acta 185 (1986) 1.
[55] R. Tibshirani, J. Royal Stat. Soc. B 58 (1996) 267.
[56] A. Smola, B. Schölkopf, Stat. Comput. 14 (2004) 199.
[57] A. Liaw, M. Wiener, R news 2 (2002) 18.
[58] L. Ward, A. Dunn, A. Faghaninia, N.E. Zimmermann, S. Bajaj, Q. Wang, J. Montoya, J. Chen, K. Bystrom, M. Dylla, Comput. Mater. Sci. 152 (2018) 60.
[59] L. Himanen, M.O. Jäger, E.V. Morooka, F.F. Canova, Y.S. Ranawat, D.Z. Gao, P. Rinke, A.S. Foster, Comput. Phys. Commun. 247 (2020), 106949.
[60] A.E. Hoerl, R.W. Kennard, Technometrics 12 (1970) 55.
[61] D.W. Scott, Multivariate density estimation: theory, practice and visualization, first ed., Wiley, 1992.
[62] K.-R. Müller, M. Finke, N. Murata, K. Schulten, S.-I. Amari, Neural Comput. 8 (1996) 1085.
[63] B. Huang, O.A. Von Lilienfeld, J. Chem. Phys. 145 (2016), 161102.
[64] M. Rupp, R. Ramakrishnan, O.A. Von Lilienfeld, J. Phys. Chem. Lett. 6 (2015) 3309.
[65] A.S. Christensen, L.A. Bratholm, F.A. Faber, A.O. von Lilienfeld, J. Chem. Phys. 152 (2020), 044107.
[66] V.L. Deringer, G. Csányi, Phys. Rev. B 95 (2017), 094203.
[67] R. Jinnouchi, K. Miwa, F. Karsai, G. Kresse, R. Asahi, J. Phys. Chem. Lett. 11 (2020) 6946.
[68] A.P. Bartók, S. De, C. Poelking, N. Bernstein, J.R. Kermode, G. Csányi, M. Ceriotti, Sci. Adv. 3 (2017), e1701816.
[69] R. Jinnouchi, R. Asahi, J. Phys. Chem. Lett. 8 (2017) 4279.
[70] R. Jinnouchi, H. Hirata, R. Asahi, J. Phys. Chem. C 121 (2017) 26397.
[71] M.O. Jäger, E.V. Morooka, F.F. Canova, L. Himanen, A.S. Foster, npj Comput. Mater. 4 (2018) 37.